

ML Experiment No. 1

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Aim

To familiarize students with Google Colab and Python libraries (NumPy, Pandas, Matplotlib and Seaborn) for loading, inspecting, cleaning, manipulating and visualizing the Titanic dataset.

Procedure

Phase 1: GitHub Repository Setup (Pre-Execution) Before writing any code, students must set up their GitHub workspace.

1. Create an account at github.com.
2. Create a new repository. • Repository Name: CSL701-Roll_No_Machine-Learning-Lab • Description: Machine Learning Lab Experiments for Semester 7 Computer Engineering. • Visibility: Public • Initialize with README.
3. Create folder: Experiments and Mini-Project.

Phase 2: Environment Setup & Data Loading

- Task 1: Import NumPy, Pandas, Matplotlib.pyplot and Seaborn.
- Task 2: Load train.csv into a Pandas DataFrame using `pd.read_csv()`.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files

uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Titanic-Dataset.csv to Titanic-Dataset.csv

```
df = pd.read_csv("Titanic-Dataset.csv")
```

Phase 3: Data Inspection

- Task 3: Display first 10 rows using `df.head(10)`.

```
df.head(10)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3734
5	6	0	3	Moran, Mr. James	male	NaN	0	0	3308
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	174

- Task 4: Display last 5 rows using `df.tail(5)`.

```
df.tail(5)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	21153
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	11205
888	889	0	3	Johnston, Miss. Catherine	female	NaN	1	2	W./t. 3513

- Task 5: Check dataset dimensions using `df.shape` (Expected: 891 x 12).

```
df.shape
```

```
(891, 12)
```

- Task 6: Display dataset information using `df.info()`.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass          891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp           891 non-null   int64
7   Parch           891 non-null   int64
8   Ticket          891 non-null   object
9   Fare            891 non-null   float64
10  Cabin           204 non-null   object
11  Embarked        889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Task 7: Generate summary statistics using `df.describe()`; record mean, minimum and maximum values of Age and Fare.

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

Task 8: Identify missing values using `df.isnull().sum()`.

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Phase 4: Data Manipulation & Cleaning

Task 9: Fill missing Age values with the median.

```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

- Task 10: Filter surviving passengers (Survived == 1).

```
survived = df[df["Survived"] == 1]
survived.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/C 31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138

- Task 11: Convert Fare column to a NumPy array.

```
fare = df["Fare"].to_numpy()  
fare
```

```
10.5167, 10.5, 7.05, 29.125, 15., 50.,  
22.45, 20, 7.75, 1)
```

- Task 12: Calculate maximum and average Fare using np.max() and np.mean().

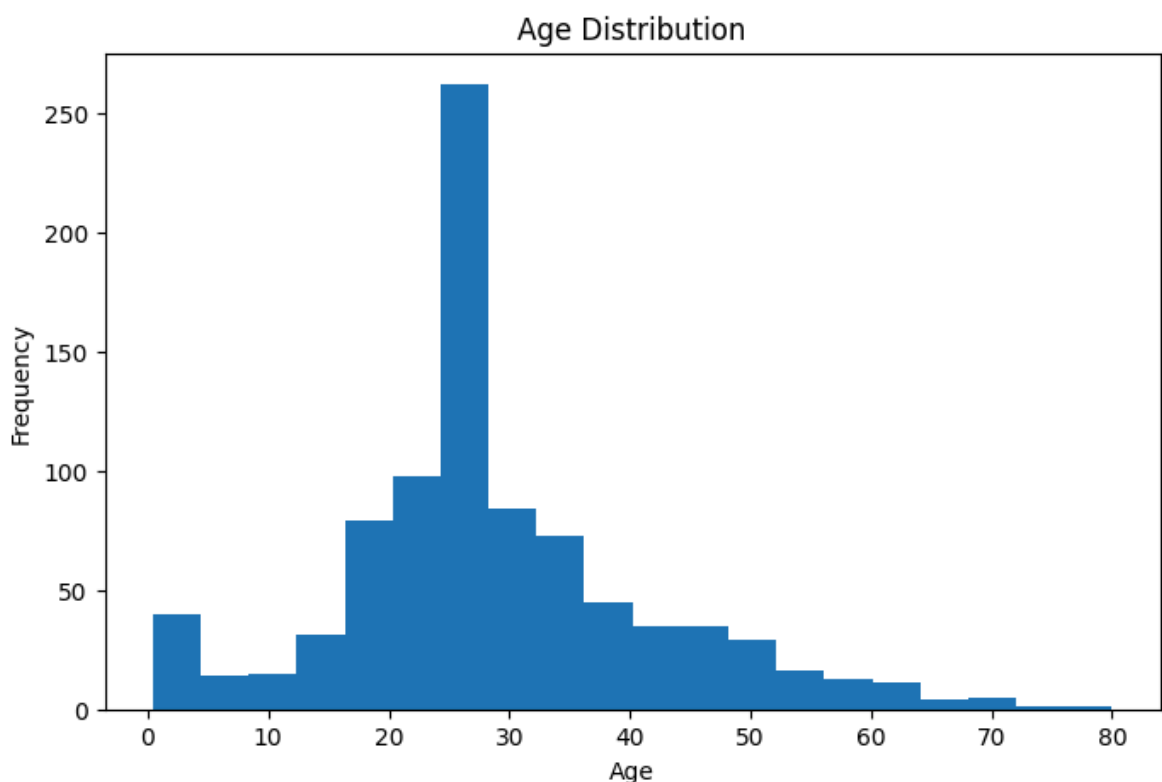
```
print("Maximum Fare:", np.max(fare))  
print("Average Fare:", np.mean(fare))
```

```
Maximum Fare: 512.3292  
Average Fare: 32.204207968574636
```

Phase 5: Data Visualization

- Task 13: Plot histogram of Age.

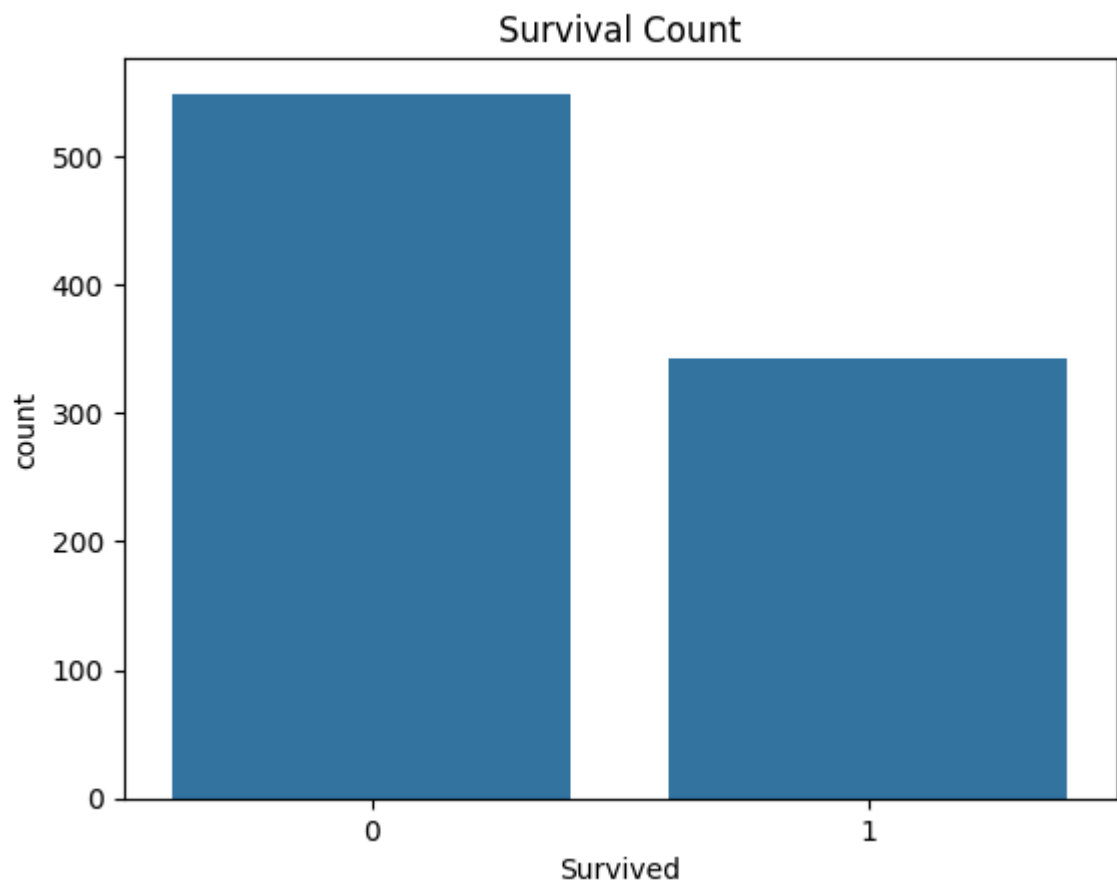
```
plt.figure(figsize=(8,5))  
plt.hist(df["Age"], bins=20)  
plt.title("Age Distribution")  
plt.xlabel("Age")  
plt.ylabel("Frequency")  
plt.show()
```



- Task 14: Plot bar chart of Survival Count.

```
sns.countplot(x="Survived", data=df)  
plt.title("Survival Count")
```

```
plt.show()
```



- Task 15: Plot scatter graph of Age vs Fare.

```
plt.figure(figsize=(8,5))  
plt.scatter(df["Age"], df["Fare"])  
plt.title("Age vs Fare")  
plt.xlabel("Age")  
plt.ylabel("Fare")  
plt.show()
```